

App Dev 2 Project Report

1.Student Details

Name: Golu kumar

Roll Number: 23f2003761

Email: 23f2003761@ds.study.iitm.ac.in

About Me: Diploma level student at IIT Madras BS in Data Science and Application, with a focus on full-stack development and building robust, data-driven web applications.

2.Project Details

Project Title: Placement Portal -v2

Problem Statement: To design a multi-role (Student/Company/Admin) web application to streamline the campus recruitment process, eliminating manual tracking errors, and providing a transparent platform for job applications, drive management, and student verification.

Approach: Built on the Flask micro-framework using an MVC architecture. It utilizes SQLAlchemy for relational data management and implements a robust permission system via custom decorators. The system features automated file lifecycle management for student resumes and real-time application tracking. Also i have used redis for caching and celery worker for cron jobs and to handle background jobs. For the authentication i have used JWT token . I have used Flask Restful for generating the API.

3.AI/LLM Declaration:

I used Gemini to assist in this project. The extent of AI/LLM usage is around 10–15%, limited to code suggestions ,documentation formatting(report writing) and some help for vue js . All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

1. Frontend

- Framework: Vue.js (3.x)
- Routing: Vue Router
- State Management: Pinia
- HTTP Client: Axios
- Styling: Bootstrap 5
- Icons: Font Awesome
- Avatars: DiceBear API

2. Backend

- Language: Python
- Web Framework: Flask
- Database ORM: Flask-SQLAlchemy
- API Structure: Flask-RESTful
- Security: Bcrypt (Password Hashing)

3. Authentication & Data

- Auth Standard: JWT (JSON Web Tokens)
- Database: Relational SQL (PostgreSQL/MySQL)

5.Database Schema / Key Tables & Relationships



6.API Resource Endpoints

Authentication

- **POST /login:** User authentication.
- **POST /register:** New user registration.
- **GET /uniqueemail:** Validate email availability.
- **GET /uniqueusername:** Validate username availability.
- **GET/PUT /profile:** Retrieve or update user profile.

Admin Endpoint

- **GET /admin/users:** Fetch all active/approved users.
- **GET /admin/unactive-users:** Fetch all pending/unapproved users.
- **PATCH /admin/approve/<user_id>:** Toggle user approval status.
- **PATCH /admin/block/<user_id>:** Toggle user blacklist status.
- **GET /admin/user/<user_id>:** Retrieve full user profile.
- **GET /admin/company/<user_id>:** Fetch company profile and active drives.
- **GET /admin/student/<user_id>:** Fetch student profile and application data.
- **GET /admin/drives:** View all placement drives.
- **GET /admin/drive/<drive_id>:** Detailed view of drive and applicant list.
- **PATCH /admin/drive/approve/<drive_id>:** Toggle drive approval status.

Company Endpoints

- **GET /company/dashboard:** Access company-specific overview.
- **POST /company/drive/create:** Register a new placement drive.
- **GET /company/drive/<drive_id>/applicants:** List all students for a specific drive.
- **GET /company/student/<applicant_id>/applications:** View specific student's history.
- **PATCH /company/application/<application_id>/status:** Update status (e.g., Selected/Rejected).

- GET /company/selected-candidates: Retrieve list of hired/selected candidates.

Student Endpoints

- GET /student/dashboard: Access student-specific overview.
- POST /student/apply/<drive_id>: Submit application for a placement drive.
- GET /student/history: View past and current applications.
- GET /student/tracking: Monitor real-time application status.
- GET /student/export-applications: Export application data reports.

7.Additional Features:

- Export user logs as CSV
- AI-generated weekly summary insights (optional extension)

8.Video Presentation

Drive link :- [link](#)